

BASIC HOSPITAL PROCEDURES

This unit specifies the competencies required to apply basic hospital procedures. It involves demonstrating the knowledge of a health facility set up, medical equipment in basic patient care identifying elements of total-patient-care and applying basic patient care procedures.

Summary of Learning Outcomes

1. Demonstrate the knowledge of a health facility set up
2. Demonstrate the knowledge of medical equipment in basic patient care
3. Identify elements of total-patient-care
4. Apply basic patient care procedures

CHAPTER 1

1 1.1 Learning Outcome 1: Demonstrate the knowledge of a health facility set up

2 1.2 Introduction to the learning outcome

This learning outcome specifies the content of competencies required to demonstrate the knowledge of a health facility set up. It comprises meaning of terms, types of departments in a health facility, Management structure and ministerial organogram, levels of health facilities, theatre layout, units in a theatre learning activities, self-assessment questions and answers, learning activities, tools, equipment, supplies and materials and references.

3 1.3. Performance Standard

1.3.1 Existing departments in a health facility are identified as per the institution's policy

1.3.2 Management structure is identified as per the MoH organogram

1.3.3 Service charter is identified as per the institutional framework

1.3.4 Theatre layout is described as per WHO standards

4 1.4 Information Sheet

1.4.1 Meaning of terms

- A *health facility* is, in general, any location where *healthcare* is provided. They include hospitals, clinics, outpatient care centers, and specialized care centers, such as birthing centers and psychiatric care centers.
- *Healthcare facility* means an office or institution providing care or treatment of diseases, whether physical, mental, or emotional, or other medical, physiological, or psychological conditions, including but not limited to, hospitals, rehabilitation hospitals or other clinics, including weight control clinics, nursing homes, long-term care facilities, homes for the aging or chronically ill, laboratories, and offices of surgeons, chiropractors, physical therapists, physicians, psychiatrists, dentists, and all specialists within these professions. This definition shall include all waiting rooms, hallways, private rooms, semiprivate rooms, and wards within health care facilities.
- A *medical procedure* is a course of action intended to achieve a result in the delivery of healthcare.
- *Medical test* is a procedure with the intention of determining, measuring, or diagnosing a patient condition or parameter.

1.4.2 Management structure and ministerial organogram

Hospitals require precision in the execution of job responsibilities and multiple layers of accountability in order to function. To accomplish this, hospitals use a vertical organizational structure with many layers of management.

Understanding hospital organizational structure ensures that hospital employees know their own responsibilities, the responsibilities of those around them, to whom they report and who to talk to about particular responsibilities or fields of knowledge.

The Organizational Structure of Hospitals

Board

All hospitals include some form of governing body responsible for making high-level decisions about the organization. A hospital's board of directors is often drawn from the healthcare community and is made up of experts in their respective fields. Religiously affiliated hospitals often include clergy on their boards of directors. Teaching hospitals often include university faculty from the medical school with which they're affiliated.

Executives

Hospital executives are responsible for managing the organization, making financial decisions and overseeing business strategy. Medical and health services managers may oversee entire practices or clinical areas. A hospital typically has a chief financial officer who tends to the financial health of the business and a chief operating officer or chief executive officer responsible for high level business strategy and decision making.

Department Administrators

Department administrators report to the hospital executives and manage the day-to-day operations of specific departments within a hospital. The chief of surgery, for example, is responsible for overseeing daily activities within the surgical department as well as performing surgery. A chief of surgery might engage in public relations activities, fundraising and recruitment. Non-medical departments within a hospital, such as food services or switchboard personnel, also have department administrators.

Patient Care Managers

Nurse managers and supervising physicians are both patient care managers. These individuals manage small groups of professionals who provide direct patient care. They ensure that orders are carried out, that hospital employees are fulfilling their duties appropriately and that employees are complying with legal requirements.

Service Providers

The vast majority of hospital workers are service providers: doctors, nurses, orderlies, physical therapists, laundry workers and the many other people required in order for a hospital to function. They provide patient care, maintain records and ensure that the hospital is able to deliver care to patients in an effective manner.

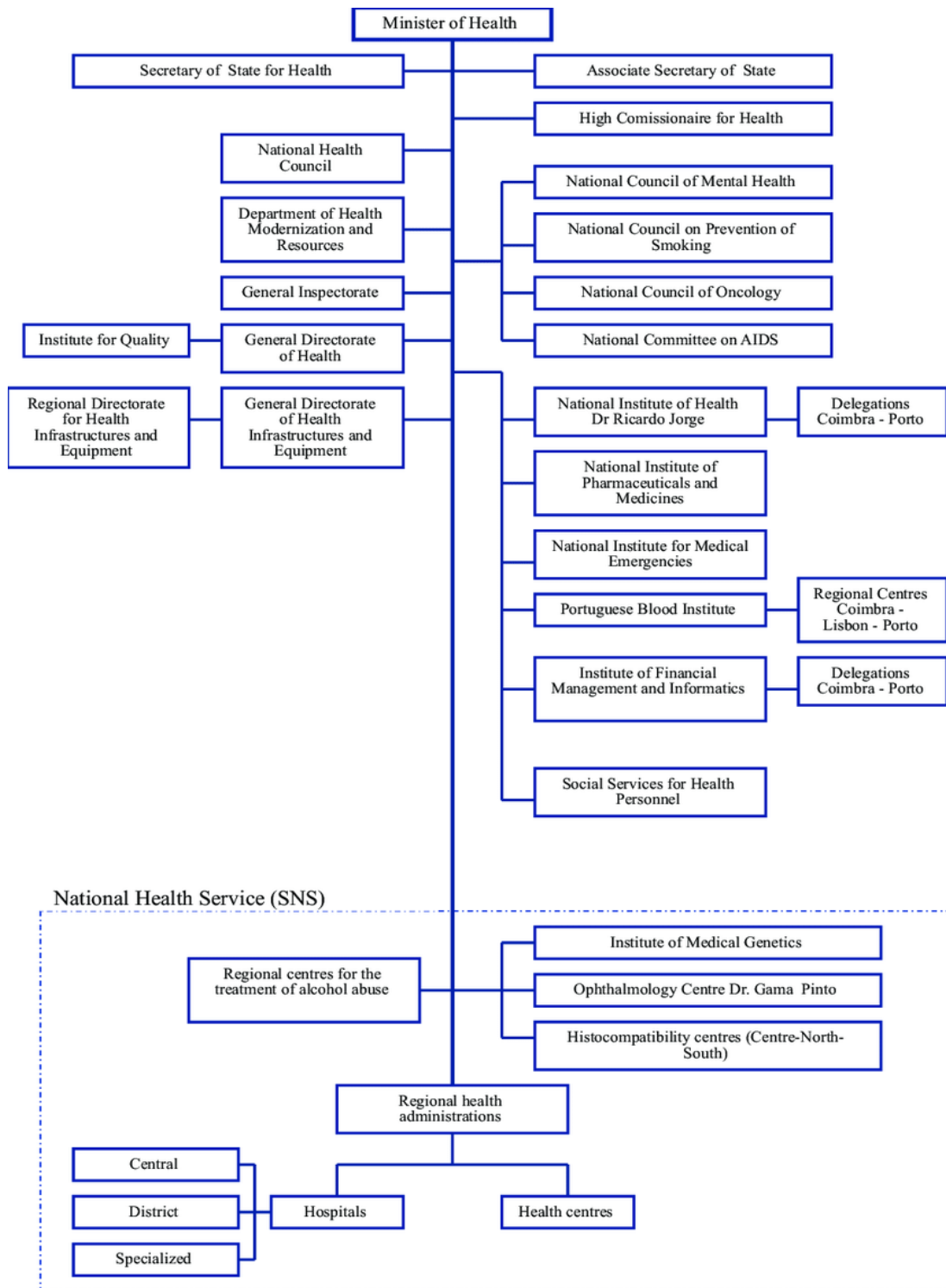


Figure 1: Organizational chart of Ministry of Health structure

Source: https://www.researchgate.net/figure/Organizational-chart-of-Ministry-of-Health-structure_fig10_237624652

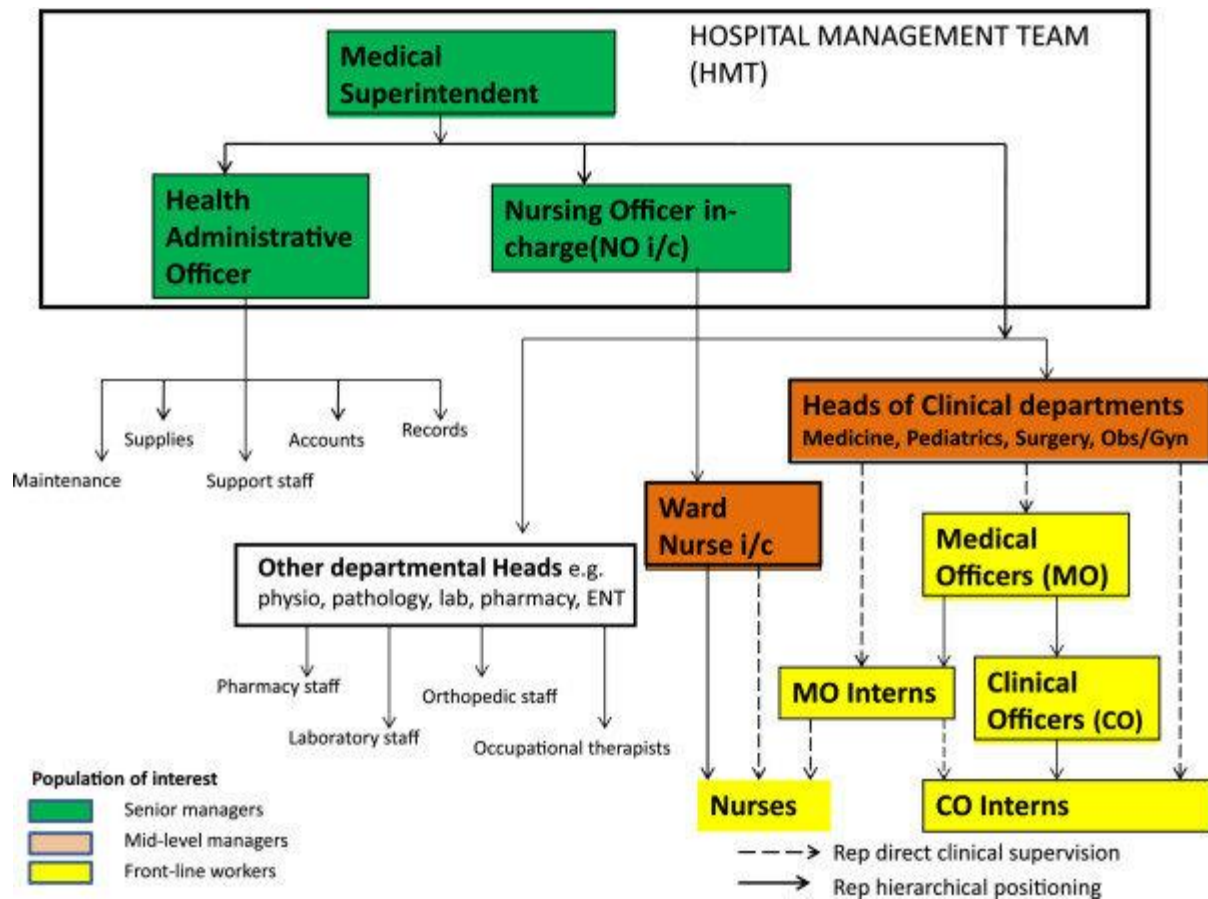


Figure 2: Generic organogram of district hospitals in Kenya. *Medical officers here refer to physicians while clinical officers are “non-physician” clinicians as described in 16

Source: https://www.researchgate.net/figure/Generic-organogram-of-district-hospitals-in-Kenya-Medical-officers-here-refer-to_fig1_235739577

1.4.3. Types of departments in a health facility

Departments in Hospital

There are several departments are found in hospitals namely Outpatient department (OPD), Inpatient Service (IP), Medical Department, Nursing Department, Paramedical Department, Physical Medicine and Rehabilitation Department, Operation Theatre Complex (OT), Pharmacy Department, Radiology Department (X-ray), Dietary Department, Non-professional Services (Business Management), Medical Record Department (MRD) and Personnel Department.

Outpatient Department (OPD)

Most hospitals now have an OPD. The advantage of OPD is that much of the investigative unit and curative work can be done there without admitting the client, thus curtailing medical expenses.

The scope of OPD includes the following:

- a. Consultation, investigation, procedures, specialty services.
- b. Preventive and promotive health care: Clinics, which include: diabetic, antenatal, postnatal and under five.
- c. Rehabilitation services (physiotherapy, occupational therapy etc)
- d. Health education
- e. Counseling

Inpatient Service (IP)

If OPD is the show window of the hospital, the IP is the heart of the hospital. The IP service provides lodging, diet and medical care. Conveniently, it can be divided into:

- a. Wards and rooms
- b. Nurses station
- c. Dietary services

Sanitary facilities and other requirements

Medical department

The medical department has within it the various clinical services. They are: medicine, surgery, gynaecology, obstetrics, paediatrics, eye, ENT, dental, orthopaedics, neurology, cardiology, psychiatry, skin, V.D., plastic surgery, nuclear medicine, infectious disease etc. medical superintendent is a doctor who has control over all medical department.

Nursing department

The nursing department is the organizational structure through which nurses provide nursing care for clients under the jurisdiction of the institution. The nursing department consists of nursing service and nursing education. The primary purpose of the nursing service is to provide comprehensive, safe, effective and well-organized nursing care through the personnel of the department. The personnel consists of nursing superintendent, assistant nursing superintendents, head nurses and staff nurses. All of these are registered nurses, other personnel who function in the nursing service department may include the auxiliary personnel nurse aids and domestics who handle the non-nursing services.

The nursing education section has the responsibility of preparing nursing students to become professional nurses. Uplifting the standard of nursing by inservice education and refresher courses etc., are included in the functions of this department. The personnel consists of principal or director of nursing education, the associate professors, assistant professors, tutors and clinical instructors.

Pharmacy Department

Pharmacy is a crucial factor in medical factor. It should be planned and organized well. The pharmacy department has the responsibility for selecting purchasing, compounding, storing and dispensing all drugs and medications. The pharmacy should be under the supervision of a registered pharmacist.

Radiology Department (X-ray Department)

The department must be located in a place where there is easy accessibility for OP and IP clients. Of the total space, the distribution for various rooms is as follows:

- a. X-ray rooms: 25%
- b. Film processing: 10%
- c. Administration: 30%
- d. Waiting area: 5%
- e. Circulation area: 30%

Sufficient number of staff should be available. Staff must be protected against radiation hazards.

This department has the following services.

- a. Radiographic examinations and their interpretations
- b. X-ray, radium, radioactive cobalt and other radioactive therapy
- c. Ultrasonography, Echocardiogram, C.T. Scan, MRI and ECG.

Dietary Department

The dietary department has the responsibility for the food service to the client according to their needs and doctor's prescription. This department is responsible for the health teaching in regard to proper diet of the client upon their discharge from the hospital.

Non-Professional Services (Business Management)

Admitting Department

The admitting department has the responsibility for admitting the client to the hospital. The importance of this department lies in the public relation that is maintained.

The client, his family and his friends must be treated with utmost respect, courtesy and tact. The enquiries made about the hospital and other clients are to be answered appropriately.

Administration

The administration of the entire hospital cannot be vested on the administrator alone. It is a collective responsibility of a group of people. The administrative staff, depending upon the size of the hospital, is composed of the administrator, the assistant administrator, the business manager and the departmental heads.

Purchasing Department

The purchasing department has the responsibility for purchasing all supplies and equipments for the hospital.

Accounts (Business Office)

This department has the responsibility for collecting the money which is owed to the hospital, paying for the supplies and equipment, handling all records pertaining to hospital finance, keeping records of assets and liabilities and assisting with budget. The business manager is responsible for the functions of the department. The accountants help him.

Housekeeping

The housekeeping department has one main function – to keep the hospital clean.

Laundry

The laundry takes care of the entire team linen of the hospital. It has the following functions:

- a. Washing the dirty linen
- b. Repairing the torn linen
- c. Replacing the condemned linen

Mechanical Department

Electricity, water supply, heat, air-conditioning etc., are looked after by the mechanical department.

Maintenance Department

The maintenance department keeps the hospital in a good state of repair. Carpenters, painters, welders, gardeners etc., are included in the personnel of this department.

Central Supply Department

The purpose of the central supply department is to prepare and furnish other departments with equipment and supplies needed in the client care e.g. syringes, needles, treatment trays.

Social Service

The social service department assists in obtaining financial aid for clients and their families. This department services also as a liaison between the client and community agencies.

Pastoral Care

Under the leadership of the chaplain, the pastoral care team meets the spiritual needs of the client.

Some departments function as a part of other departments already mentioned, e.g., the operating room functions as part of the department of surgery. The outpatient department is a combination of several departments. The emergency room functions along with the department of medicine. Sometimes according to the load of client care, the services may be given in special department such as intensive care, immediate care, and ambulatory care units.

Hospital Waste Management

It is newly set department which takes care of the disposal of the entire waste both solid and liquid.

Central Sterile Supply Department (CSSD)

This is important department which supplies sterile articles throughout the hospital. CSSD handles contaminated, clean and sterile articles.

Work flow in CSSD: Receiving – Washing – Drying – Accounting – Sorting – Packing – Sterilization – Sterile storage – Issue. The articles should move in one direction from receipt to issue. The location should be such that the wards and departments can have easy access.

Medical Record Department (MRD)

This is an integral part of every modern hospital. The guiding principle is “people forget, records remember.” Functionally the MRD is divided into

- (1) Reception
- (2) Medical Records Library
- (3) Statistical Section.

Weeding out of clients file, is done in successive years:

- a. OP Records: 5 years.
- b. IP Records: 10 years.
- c. Medicolegal Records: 15 years.

Now with computerization the files can be entered into the computer and can be utilized when required.

Every health care facility should have arrangement for handling medicolegal cases. The hospital administration, as well as the doctors, nurses and other staff members should be made

aware of the legal implications involved in the client's care, so that lot of problems can be avoided.

Personnel Department

This department in the hospital must be well versed with law of the land especially the labour laws and is responsible for recruitment, selection, promotion, transfer, termination etc. The personnel department functions under the personnel officer who is qualified in the personnel administration. The personnel department has the following functions, directed to the welfare of the personnel.

- a. Recruitment of personnel
- b. Interviewing prospective employees
- c. Promotion and transfer of employees
- d. Termination of employment
- e. Inservice training programme
- f. Remuneration and incentives
- g. Safety
- h. Health programme
- i. Recreation

Paramedical departments

Paramedical departments are adjunctive to the practice of medicine in the maintenance or restoration of health and normal functioning.

They include:

Pathology Department

The following laboratories are usually found in the pathology department:

- Bacteriology laboratory: This laboratory studies about the bacteria and their toxins.
- Biochemistry: this is concerned with the chemistry of living organisms and of vital process.
- Haematology laboratory: it is responsible for making haemoglobin determinations, coagulation time studies, red and white cell counts and special blood pathology studies for anaemia and leukaemia etc.
- Parasitology laboratory: it studies the presence of parasites, the cyst and ovas of the parasites that are found in the faeces.
- Serology laboratory: it does blood agglutination tests, Wassermann tests, V.D.R.L. etc.

- Blood bank: it has the responsibility for collecting and processing all blood used in the hospital for transfusions. It makes studies on newborn infants who may have haemolytic diseases and does antibody studies on the prenatal client.
- Histopathology department: it prepares tissues for gross and microscopic studies.
- Laboratory services (LAB) must be available day and night. Must be located on the ground floor and should be easily accessible to the outpatients. Space requirement of Lab is:
- Primary space: Required for technical work.
- Secondary space: space utilized for administrative purpose.
- Circulation space: for unchattered movement of personnel and equipment.
- There should be sufficient staff and work arrangement for the efficient functioning of the department.

Physical Medicine and Rehabilitation Department

This department deals with clients who have functional disabilities resulting from disease conditions/injuries. This department can have physiotherapy, occupational therapy, speech therapy and vocational training. This department will be under the direction of a well – qualified physician who has special training in the field of physical medicine and rehabilitation. His staff should include therapists with qualification in the various specialties.

Operation Theatre Complex (OT)

This consists of one or more operation theatres and other facilities. OT complex must be located in a place where there is easy and quick access to the delivery suite. These should be four zone – outer zone, clean zone, sterile zone, disposal zone. There should be a sterilization room with an autoclave. The number of OT depends on many factors. There should be an arrangement for good lighting and ventilation.

Delivery suite is the place where births take place. The delivery suite is divided into three zones are first stage room, second stage room, delivery room. The room should have good lighting and ventilation. It should have adequate number of staff.

1.4.4 Levels of health facilities

Kenya's Health sector is one of the 14 devolved functions managed by the 47 county governments as provided in the Fourth Schedule of the 2010 Constitution.

County health facilities and pharmacies, ambulance services, promoting primary health care, licensing and controlling facilities that sell food to the public and veterinary services are some of the health-related roles the 47 counties manage. County health facilities and services

include county referral hospitals, sub-county health facilities, environmental health services, communicable disease control, nutrition, family planning, maternal and child health plus Health Education.

There are six different levels of health care facilities. The first five are managed on the county level, the sixth level by the national government. In this system the patients may move from one level to the next by using a referral letter.

LEVEL 1 – Community Facilities

They are run by certified medical clinical officers.

Some of the services:

- Treatment of minor ailments like diarrhoea
- Tuberculosis (TB) screening, home visits, contact tracing of TB patients and tracing of TB defaulters
- Screening of malnutrition
- Malaria rapid test
- Blood pressure and blood sugar testing
- HIV testing
- Health talks with pregnant women and observations of signs of danger
- Issuance of referral letters to other facilities

LEVEL 2 – Health Dispensaries

These facilities are run by clinical officers:

The dispensaries in the cities act like a health centre (see level 3), with the difference that the dispensary does not have in-patient facilities.

These are some of the services you will expect in a dispensary:

- Outpatient services
- VCT services
- Tuberculosis services
- Laboratory Services
- Well baby Clinics

- Antenatal and Postnatal services
- Pharmacy
- Counselling services
- Curative treatment
- They issue referral letters to other facilities

LEVEL 3 – Health Centres

These are small hospitals with minimal facilities, yet they offer services like the big hospitals.

They are run by at least one doctor, clinical officers and nurses.

These are some of the services they offer:

- Maternity in-patient services with a ward
- Curative services
- Laboratory services
- Dental
- Counselling
- Pharmacy
- TB Clinics
- Diabetes & hypertension clinics
- Comprehensive care clinics for patients living with HIV
- Baby well clinics
- Antenatal and postnatal services
- They issue referral letters to other facilities

LEVEL 4 – County Hospitals

- These are hospitals that offer holistic services and are ran by a director who is a medic and at best a doctor by profession
- In many counties there's just one hospital but in larger cities like Nairobi there are two
- They have in principle the same services as the Level 3 hospitals, plus X-Ray services
They issue referral letters to other facilities

LEVEL 5 – County Referral Hospitals

These are the county referral hospitals formerly the provincial hospitals. They are run by Chief Executive Officers who are medic by profession and have over 100 beds capacity for their in-patient. They are also do research about health.

In Nairobi Mama Lucy Hospital and Mbagathi Hospital both double up as county referral hospitals and Level 4 hospitals.

Services include what other hospitals offer, plus

- Ultrasound
- CT-Scan
- Surgery
- Pharmacy
- Physiotherapy
- Orthopaedics
- Occupational Therapy
- They issue referral letters to other facilities

LEVEL 6 – National Referral Hospitals

In Kenya there are three Teaching and Research referral hospitals: Mathari Hospital, Kenyatta National Hospital, Moi Teaching and Referral Hospital and the National Spinal Injury Referral Hospital. Their range of services is the same as of on Level 5, but they offer specialised treatments to patients and are not only accessed by Kenyans but do serve East Africa and Central Africa.

- a. Mathari Teaching and Referral Hospital offers specialised mental services.
- b. Kenyatta National Hospital and Moi Teaching and Referral hospital offer specialised consultations in curative care.
- c. National Spinal Injury Referral offers specialised services in orthopaedic and spinal injuries.

The national government manages these three hospitals.

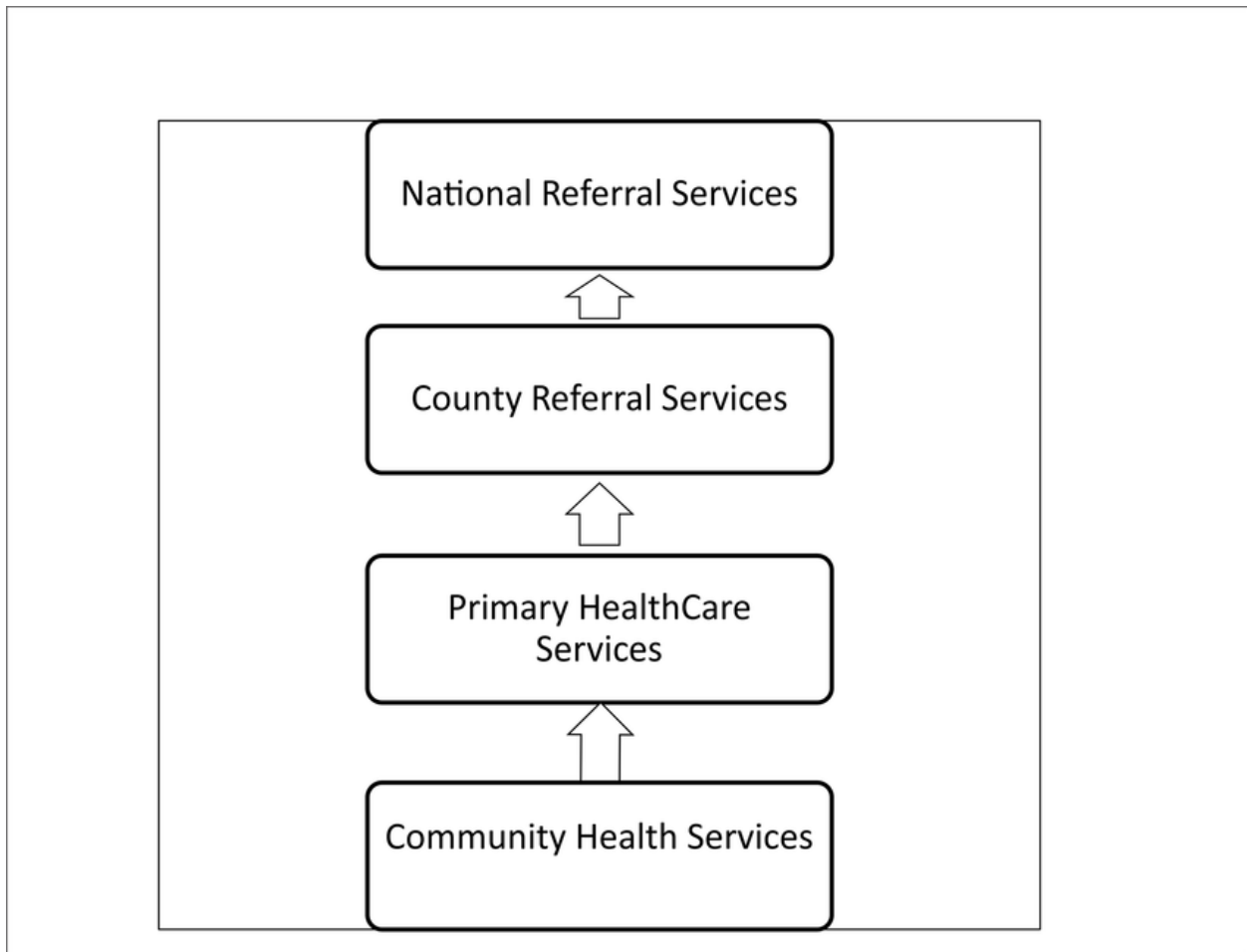


Figure 3: Organization of health services in Kenya under a devolved system.

Source: https://www.researchgate.net/figure/Organization-of-health-services-in-Kenya-under-a-devolved-system_fig5_287964619

1.4.5 Theatre layout and units

Physical layout

a. Arrangement.

Basic principles for efficient workflow are adhered to when planning the arrangement of rooms within a surgical suite. For example, traffic must move smoothly and without interruption into, through, and out of the suite. Those rooms where support activities are performed are grouped conveniently and systematically about the ORs, which are arranged in pairs with scrub rooms and sterilizing facilities between them as shown in **Figure 1**. Proper

arrangement of the rooms reduces the flow of excess traffic, saves much unnecessary walking, and therefore conserves time and effort.

b. Size of the Suite.

Numerous factors are considered when estimating the size of the surgical suite required for a hospital:

1. The first of these is the operative load or the number of surgical cases to be done per day. Some hospitals have more patients requiring medical care than surgical care. In such a hospital, only a small operative section would be necessary. Several operative procedures can be done per day in each OR, one case following another, but all should be completed by the end of the usual day shift or shortly thereafter.
 - i. The average number of ORs needed is indicated by the number of operative cases to be done daily divided by the number of cases that can be done daily in one OR.
 - ii. Operative load can also be broadly interpreted to include the amount of work to be done per day, outside of actual cases. There is a great deal of preparatory work necessary for each surgical procedure as well as the routine tasks to maintain the suite. If most supplies are prepared and sterilized in CMS, the OR suite needs to maintain only a small preparation and sterilization area; otherwise, it may need a large one.
2. The second factor, the types of surgery to be done, is even more specific. Some procedures such as cardiac surgery are very long and detailed, requiring many hours of operative time; others, such as tonsillectomy, can be accomplished in just 20 to 30 minutes. In the first instance, a single OR would be engaged for the whole operative day with just one case; in the second example, eight or ten such procedures could be done in the same length of time.
3. The number and frequency of emergency cases are also considerations related to the necessary size of the suite. The larger the troop area near a hospital, the greater the number of accident and emergency cases that may have to be done. These must be done immediately, meaning that other scheduled cases may have to wait until after normal duty hours, unless provision has been made for an additional room.

c. Size of Individual Operating Room.

In addition to plans for the number of ORs required, the size of each room must also be considered. Certain types of surgery require the use of much bulky equipment. This is especially true of open-heart surgery, during which large machines are required to pump and oxygenate the blood outside of the body and to monitor the functioning of the heart. Factors such as this must be planned for and sufficient floor space provided so that breaks in aseptic technique due to overcrowding will be avoided. On the other hand, rooms should be no larger than necessary, thus reducing the amount of unnecessary times spent walking. The Army criteria for square footage per room are sufficiently flexible to provide for the needs of its hospitals as shown in Figure 1, "Type" floor plan for an Army hospital-operating suite.)

d. Four Areas.

The designations listed in the Four Zone Concept (1) thru (4) are not necessarily used for all hospitals, but whenever feasible the surgical suite is segregated into four areas for traffic control. The purpose of such control is to assure maximum protection against infections. In analyzing the traffic and commerce system of the OR system, specific traffic patterns must be determined. These are dependent on the entrances and exits for both personnel and materials. Renovation planning of existing facilities should consider renovation of central supply and storage areas to bring these as close to the point of utilization as possible. Where entirely new wings, buildings, or entire hospital complexes are being considered, there is opportunity to design traffic, materials-handling, and storage systems around the requirements of the surgical suite. Traffic control design is aided by designation of the four-zone concept (as shown below): the interchange area, semirestricted area, restricted area, and dirty area.

INTERCHANGE	SEMI-RESTRICTED	RESTRICTED	DIRTY
Patient reception	Induction rooms	Operating rooms	Disposal Area
Locker rooms	Clean storage rooms	Sterile supply storage areas	
Lounges	Scrub areas		
Offices	Recovery rooms		

Figure 4: Medical Theatre layout

Four-zone concept.

1. Interchange area.

The interchange area is provided to enable persons from outside the surgical suite to meet with personnel within the suite when necessary. Examples of such rooms are the offices--one for the OR supervisor and one for the chief of anesthesiology as shown in Figure 1 whenever the offices for these personnel are situated within the surgical suite. In some hospitals, these offices are located convenient to but outside the surgical suite in order to keep traffic within the suite to a minimum. If there is a postanesthesia (recovery) room, it is also classified as an interchange area although entry is carefully controlled. The main outer corridors are in the interchange area. This area also consists of the rooms designed for personnel to change from street clothes or military uniforms into OR attire ("scrub clothing") as shown Figure 1. If space and arrangement permit, there are two doors to each dressing room--one for the entry of persons in street clothes and the other for their exit in scrub clothes. The individual lockers are generally equipped with nametags, locks, and keys. Caps, masks, and scrub clothing are stored in the dressing rooms. Dressing rooms may be equipped with showers. A mask need not be worn in this area.

2. Semirestricted area.

Comprising the semirestricted area are corridors within the work area, the anesthesia workroom and storage, the utility closet, the instrument storage room, the workroom, the linen storage room, the clean linen room, and the soiled linen room as shown in Figure 1. The semirestricted area then represents the supply center for the surgical suite. Although, non-sterile procedures are performed in this area, personnel without OR attire are not admitted. A mask need not be worn in this area.

3. Restricted area.

The restricted area consists of all rooms in which sterile procedures are done and sterile goods are opened or exposed. This includes the individual ORs and the adjacent scrub rooms where soap, nail files, brushes, timers, and written directions are available for the scrub procedure. Also included are the preanesthesia rooms (induction rooms) if the suite has this facility. On the floor plan shown in figure 1-3, a part of the "anesthesia area," (4), may be designated for the induction of anesthesia. If the suite is arranged to include a substerilizing room between each two ORs, then these rooms are considered to be restricted because supplies are autoclaved

here without being covered and are moved, uncovered, into the OR. Each person working in the restricted area is required to wear scrub clothes, cap, and mask.

4. Dirty area. The dirty area is the disposal area, where all utilized materials and linen are gathered, packaged, and sent to appropriate areas.

e. **Arrangement of Areas.**

There is no one rule that must be followed in the physical arrangement of the four areas previously discussed. However, the nonrestricted and interchange areas should be located near the entry door. Workroom areas are situated near the center of the suite, and storage and supply rooms nearby are positioned to avoid waste in time and energy of personnel. Areas for storage of both sterile and unsterile supplies must be clearly marked to avoid mistaking one for the other. If there is no linen chute in the suite, a room for soiled linen is necessary. This room is classified as semirestricted. If the suite has observation galleries, these are provided with outside entries to eliminate unnecessary traffic of persons in street clothing.

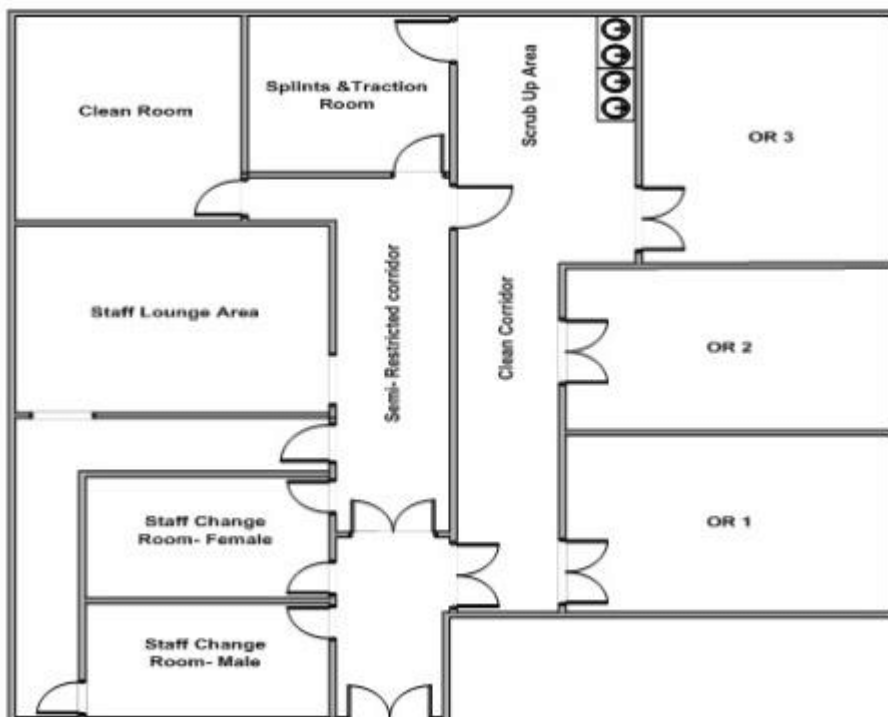


Figure 5: Theatre layout

Source: https://www.researchgate.net/figure/First-case-modified-OT-layout-plan_fig2_256457508

5 1.5 Learning Activities

1.5.1 Practical activity

Field visit:

Conduct a visit to the nearest county health facility and do the following:

- a. Classify it based on level.
- b. Assess the compliance of the layout and draw a sketch.
- c. Draw the organization structure of the facility

1.5.2 Self-Assessment Questions

Explain four zone concept in theatre layout.

Describe Level 3 health facilities and services offered.

Describe nursing department

1.5.3 Answers to Self-Assessment questions

Explain four zone concept in theatre layout.

Four-zone concept.

1. Interchange area.

The interchange area is provided to enable persons from outside the surgical suite to meet with personnel within the suite when necessary. Examples of such rooms are the offices--one for the OR supervisor and one for the chief of anesthesiology as shown in Figure 1 whenever the offices for these personnel are situated within the surgical suite. In some hospitals, these offices are located convenient to but outside the surgical suite in order to keep traffic within the suite to a minimum. If there is a postanesthesia (recovery) room, it is also classified as an interchange area although entry is carefully controlled. The main outer corridors are in the interchange area. This area also consists of the rooms designed for personnel to change from street clothes or military uniforms into OR attire ("scrub clothing") as shown Figure 1. If space and arrangement permit, there are two doors to each dressing room--one for the entry of persons in street clothes and the other for their exit in scrub clothes. The individual lockers are generally equipped with nametags, locks, and keys. Caps, masks, and scrub clothing are stored in the dressing rooms. Dressing rooms may be equipped with showers. A mask need not be worn in this area.

2. Semirestricted area.

Comprising the semirestricted area are corridors within the work area, the anesthesia workroom and storage, the utility closet, the instrument storage room, the workroom, the linen storage room, the clean linen room, and the soiled linen room as shown in Figure 1. The semirestricted area then represents the supply center for the surgical suite. Although, non-sterile procedures are performed in this area, personnel without OR attire are not admitted. A mask need not be worn in this area.

3. Restricted area.

The restricted area consists of all rooms in which sterile procedures are done and sterile goods are opened or exposed. This includes the individual ORs and the adjacent scrub rooms where soap, nail files, brushes, timers, and written directions are available for the scrub procedure. Also included are the preanesthesia rooms (induction rooms) if the suite has this facility. On the floor plan shown in figure 1-3, a part of the "anesthesia area," (4), may be designated for the induction of anesthesia. If the suite is arranged to include a substerilizing room between each two ORs, then these rooms are considered to be restricted because supplies are autoclaved here without being covered and are moved, uncovered, into the OR. Each person working in the restricted area is required to wear scrub clothes, cap, and mask.

4. Dirty area. The dirty area is the disposal area, where all utilized materials and linen are gathered, packaged, and sent to appropriate areas.

Describe Level 3 health facilities and services offered.

LEVEL 3 – Health Centres

These are small hospitals with minimal facilities, yet they offer services like the big hospitals. They are run by at least one doctor, clinical officers and nurses.

These are some of the services they offer:

- Maternity in-patient services with a ward
- Curative services
- Laboratory services
- Dental
- Counselling

- Pharmacy
- TB Clinics
- Diabetes & hypertension clinics
- Comprehensive care clinics for patients living with HIV
- Baby well clinics
- Antenatal and postnatal services
- They issue referral letters to other facilities

Describe nursing department

The nursing department is the organizational structure through which nurses provide nursing care for clients under the jurisdiction of the institution. The nursing department consists of nursing service and nursing education. The primary purpose of the nursing service is to provide comprehensive, safe, effective and well-organized nursing care through the personnel of the department. The personnel consists of nursing superintendent, assistant nursing superintendents, head nurses and staff nurses. All of these are registered nurses, other personnel who function in the nursing service department may include the auxiliary personnel nurse aids and domestics who handle the non-nursing services.

The nursing education section has the responsibility of preparing nursing students to become professional nurses. Uplifting the standard of nursing by inservice education and refresher courses etc., are included in the functions of this department. The personnel consists of principal or director of nursing education, the associate professors, assistant professors, tutors and clinical instructors.

6 1.6 Tools, Equipment, Supplies and Materials

The following resources are provided:

Tools

- PPEs
 - ✓ Safety boots
 - ✓ Goggles
 - ✓ Dust coats
 - ✓ First aid kit

✓ Overalls

- Stationery
- Cupboard drawers
- Internet

7 1.7 References

<https://www.canestar.com/list-of-departments-in-hospital-and-its-functions----a-simple-learning-for-nurses..html> 25/02/2021 at 13:34pm

Mwabu, G., Ainsworth, M., & Nyamete, A. (1993). Quality of medical care and choice of medical treatment in Kenya: an empirical analysis. *Journal of Human Resources*, 838-862.

<https://roggkenya.org/about-this-site-reporting-on-good-governance-and-corruption-in-kenya/>

8 2.1 Learning Outcome 2: Demonstrate the knowledge of medical equipment in basic patient care

9 2.2 Introduction to the learning outcome

This learning outcome specifies the content of competencies required to demonstrate the knowledge of medical equipment in basic patient care. It comprises meaning of terms, types of health facility equipment and instruments, functions of health facility equipment and instruments, care and maintenance of health facility equipment and instruments, learning activities, self-assessment questions and answers, learning activities, tools, equipment, supplies and materials and references.

10 2.3. Performance Standard

2.3.1 Health facility equipment and instruments are identified as per the workplace procedures

2.3.2 Functions of health facility equipment and instruments are identified as per the workplace procedures

2.3.3 Malfunctioning equipment and instruments are identified and reported as per the workplace procedures

11 2.4 Information sheet

2.4.1 Meaning of terms

- *Radiology* is the science dealing with X-rays and other high-energy radiation, especially the use of such radiation for the diagnosis and treatment of disease.
- *Diagnostic* is concerned with the diagnosis of illness or other problems.
- *Magnetic resonance imaging (MRI)* is a medical *imaging* technique used in radiology to form pictures of the anatomy and the physiological processes of the body. *MRI* scanners use strong *magnetic* fields, *magnetic* field gradients, and radio waves to generate images of the organs in the body.
- *Cardiopulmonary bypass (CPB)* is a technique in which a machine temporarily takes over the function of the heart and lungs during surgery, maintaining the circulation of blood and the oxygen content of the patient's body.
- A *nebulizer* is a piece of medical equipment that a person with asthma or another respiratory condition can use to administer medication directly and quickly to the

lungs. A *nebulizer* turns liquid medicine into a very fine mist that a person can inhale through a face mask or mouthpiece.

2.4.2 Types of health facility equipment/instruments and functions

Diagnostic Equipment

Diagnostic medical equipment is any type of equipment or tools used in a hospital setting for the sole purpose of diagnosing a patient's condition. Based on the symptoms described by the patient, a diagnostic test is performed using the appropriate equipment to evaluate the patient internally. The doctor or technician is looking for any abnormalities in the affected organs or parts of the body that are causing the symptoms to be exhibited.

Below are some of the most common types of diagnostic equipment used in laboratories and medical clinics:

- a. Medical imaging machines – Medical imaging is a type of technology that is used to create visual representations of the human body's interior. The visual image produced by the equipment is used for clinical analysis and medical intervention. There are many medical imaging equipment, such as radiography (X-ray machine), computed tomography (CT scan), magnetic resonance imaging (MRI scan), ultrasound, and echocardiography, to name a few.
- b. Aside from medical imaging machines, there are other medical devices used for diagnosing patients. Some examples include patient scales, stethoscopes, dopplers, and pulse oximetry.

Durable Medical Equipment (DME)

This type of medical equipment is used mainly for providing therapeutic benefits for certain conditions or illnesses. The use of this equipment must be prescribed by a physician, which is designed to serve a medical purpose. It is a long-term and reusable device that can be used in the hospital or at home for patient care. There are several types of durable medical equipment such as the following:

- Manual wheelchairs or electric wheelchairs
- Hospital beds
- Walkers, crutches, canes (or any similar type of mobility assistive equipment)
- Traction equipment
- Pressure mattresses
- Insulin pumps

- Breast pumps
- Nebulizers
- Patient lifts
- Bili lights and blankets
- Kidney machines
- Stationary or portable oxygen tanks
- Oxygen Concentrators Ventilators

The term ‘durable’ is derived from the fact that these types of equipment have been tested for quality. They are designed to provide support for the safety and comfort of the patients. They should be able to deliver just that. Some of the common features of durable medical equipment include non-slip features and load-bearing strength. Since this equipment is prescribed by the physician for the benefit of the patient, the cost of acquiring one is often covered by health insurance plans.

Treatment Equipment

Treatment equipment is any type of medical device or tool that is designed to treat a specific condition. It utilizes modern technology in order to address any abnormalities to restore function in the affected organs or tissues within the body. This can also include the surgical supplies designed to provide treatment for certain conditions that require surgical intervention.

Below are common examples of medical treatment equipment that you will find in hospitals and clinics:

- **Infusion Pumps** – This type of treatment equipment is used in a hospital setting. It is designed to infuse medication, fluids, and other forms of treatment to the patient’s circulatory system. This machine is used intravenously but you can also find epidural or arterial infusions. This machine is reliable because it injects controlled amounts of fluids into the patient’s system over a given period of time.
- **LASIK Surgical Machines** – The use of LASIK technology is used primarily for the treatment of eye conditions. It is designed for use on patients suffering from myopia, hyperopia, or astigmatism.
- **Medical Lasers** – The laser is a revolutionary technology introduced in the medical field for the treatment of various medical conditions. It is a device that emits a wavelength of electromagnetic radiation for clinical applications. These wavelengths vary when it

comes to energy level and pulse duration. These settings are to be determined by the attending physician during treatment.

Meanwhile, treatment equipment used for surgical procedures include the following:

- Diagnostic scopes
- Surgical caps, gowns, gloves, or glasses (for the surgical team)
- Gauze and drapes (for patients)

Life Support Equipment

As the name implies, life support equipment are those medical devices intended to maintain the bodily function of a patient. Without life support, it will be difficult for the patient's organ systems to function on their own. There are a variety of life support machines in the medical field today, such as the following:

- Heart-lung Machines – Also known as cardiopulmonary bypass (CB) devices, this type of equipment temporarily functions as the heart or lungs of a patient during surgery. It facilitates in the circulation of blood and oxygen throughout the patient's body. It is known as a form of extracorporeal circulation.
- Medical Ventilators – This type of device is designed to move breathable air in and out of the lungs. It is used on patients who have difficulty breathing, or who are incapable of delivering an adequate supply of oxygen throughout the body. There are two types of medical ventilators. One is hand-operated using a bag valve mask. The other is run and operated by a computer. You will find this machine used in intensive care unit patients or home care patients.
- Dialysis Machine – Dialysis is the method of removing excess solutes or toxins from the blood. It is used on people whose kidneys have been damaged and are incapable of performing these functions. It is a type of renal replacement therapy. Dialysis machines are essential for those who have kidney damage or lost kidney function.
- Incubators – An incubator is a medical apparatus used for neonatal intensive care. It is designed to mimic the environmental conditions suitable for newborn babies. It is often used in conjunction with a few other medical equipment such as an oxygen hood and ventilator.

Medical Laboratory Equipment

The use of medical laboratory equipment is often seen in medical clinics or diagnostic laboratories. These types of equipment are intended for the analysis of blood, urine, genes, and

other biological material. Below are the most common examples of medical laboratory equipment used in the medical field:

- Blood gas analyzers
- Chemistry analyzers
- Blood collection supplies
- Electrolyte analyzers
- Differential counters
- Drug testing analyzers
- Coagulation analyzers
- Hematology analyzers
- Urinalysis analyzers
- Microbiological systems

As medical technology continues to develop, you can expect that there will be more types of medical equipment emerging. This is good news to patients who can expect quality medical care and diagnosis utilizing the latest in medical technology.

Ceiling Mounted Equipment

Expect a Surgical and an Anaesthetic Pendant in any theatre which have power, data and various gas outlets. There are several major suppliers on the market with numerous different types. Pendants can be rigid, rigid-retractable or fully articulated. Theatres require fully articulated pendants for maximum flexibility. A lot of co-ordination is required between medical gas, electrical, and pendant trade contractors together with electrical, mechanical and structural engineers/consultants while installing pendants to ensure all structural steel support is in place, gas pipes are properly connected and power and data cables run at the correct programmed dates. Various life saving equipment must be powered off Isolated Power Supply (IPS) which is also backed up by Un-interruptible Power Supply (UPS), in case of power failure during surgery.

1. Theatre Lights

There used to be the Gas Discharge lights or Halogen lights. Aspects to consider were bulb life, costs associated and Theatre down time while bulbs were changed.

There is a new kid on the block in the last few years – LED technology. About 30% more expensive compared to the existing technology but very long life bulbs (over 20,000 hours), ease of maintenance (couple of minutes to replace an LED), cheap cost of replacement (fraction of older technology) and ability to vary light temperature hence helping to diagnose cancerous cells etc. Theatre Lights must be backed up by a battery back up in case of power failure during surgery. Suggested time can be 3 hours back up. Note that general lighting and emergency escape lighting should also be on similar battery/UPS back up time. The battery backup unit can be a large wall mounted cabinet about 1m x 1m and about 300-400mm deep. It will need to be mounted not far from the Theatre Lights but not inside the clinical space.

2. Camera

In a teaching facility, a camera (now a days High Definition Camera) and microphone is also required for one way video (from Operating Theatre to Lecture Theatre and/or Seminar Rooms) and two-way audio for surgeons and students to communicate. The camera can either be installed in the handle of the main Theatre Light or installed on a separate ceiling mounted arm. Consider all implications for power and data transfer (HD requires much higher bit rate transfer). Consider a wireless Reality TV/Big Brother style microphone on the surgeon to allow freedom of movement.

In Orthopaedic Theatres you would need to consider the largest ceiling mounted item – The Laminar Flow or the Ultra Clean Ventilation (UCV) Canopy. This item will need a separate article as we are planning a general Theatre in this article.

Wall Mounted Equipment

1. Surgeons Panel

The panels can be the older style steel type or the more current Membrane Type panels which allow ease of cleaning/disinfection. The membrane can be made anti-microbial by inclusion of silver nitrate. A newer version of panels can be touch screen which is now proving very popular taking over the mechanical switch panels and allows a lot of functionality simply by programming as the Theatre changes over time and new items are added and old removed.

For ease of cleaning and aesthetics, the panels should be flush mounted.

Note that all the pendants, theatre lights, general lights, gas alarm panel, IPS/UPS, and warning signs for X-Ray in-use / Laser in-use signs outside Theatre, air sampling ducts, clocks

etc need considered and carefully co-ordinated among the trade contractors and design consultants for services and wiring.

Floor Mounted Equipment

a. Theatre Table

Generally these are rechargeable and don't necessarily require power socket close by.

b. Trolleys

Trolleys are used for instruments and equipment such as video endoscopes and anaesthetic equipment.

We mentioned ease of cleaning at the start. To enable this the walls should have special plastic enamel paint to allow chemical cleaning if required in case of disinfection. The ceiling is generally constructed of plaster board or special metal to ensure it is air-tight and easily cleaned.

Airflow regime is an important part of moving the air from clean areas towards dirty and out of the Operating Theatre suite to ensure the Theatre is the most clean environment for operating on the patients.

I trust the above has given you some basics to consider when planning your next operating theatre.

2.4.3 Care and maintenance of health facility equipment and instruments

Most commonly used electric equipments in the hospitals are:

- Computers
- Refrigerators
- Cardiac monitors
- Pulse oxymeter
- Suction apparatus
- Nebulizer
- Syringe pump
- Glucometer
- Ultrasound
- CT & MRI machines
- Ventilator
- Defibrillator

Proper care and maintenance is very important for proper functioning of equipment.

Medical Equipment Care and Maintenance Tips

There's just no getting around the fact that medical equipment is expensive. Even if you manage to find gently used equipment that is in good condition and current enough to benefit your practice, you'll probably still shell out some dough for it. So you want to make sure it lasts as long as possible.

This requires you to practice proper care and maintenance, especially for the equipment you tend to use day in and day out. You need it to work properly so that you can offer your patients the best possible care. If it breaks down, not only are you left trying to figure out how to treat patients on the fly, but it doesn't exactly make your practice look reliable.

Even with proper maintenance and repair, you can't always control equipment failures, but your best bet to keep these pricy machines in good working order is consistent care. Here are a few maintenance and care tips to get you started.

1. Proper Usage

Maintaining your medical equipment starts with the way you treat it every day, including how you use it. Smacking your old tube TV may have improved reception, but it's definitely not going to make your sensitive equipment work better or faster.

Okay, so you're probably not going to hit your machinery, but if you're using it improperly, you might as well. Your mantra should be "use only as directed", and with most machinery, you'll have plenty of in-depth instructions to peruse. If you're not sure about proper usage, take the time to call customer support with any questions.

When you use equipment in a way that is not recommended, you not only risk harming the machinery (and possibly patients), but you may open yourself up to liability. In other words, it's important to practice proper usage.

2. Follow Manufacturer Guidelines

Manufacturers not only provide instructions related to how to properly use equipment, but also how to maintain it. Whether you're supposed to run regular diagnostics and updates to firmware and software or you're advised to hire professional technicians for regular inspection and maintenance, it's always best to follow manufacturer guidelines concerning maintenance.

This will give you the best opportunity to ensure proper function and longevity, as well as help you to avoid potentially costly snafus should machinery fail.

3. In-house Maintenance

There are certain maintenance and repair tasks for which your staff probably isn't qualified. However, you may find that you, your office personnel, or in-house support services (if you're in a hospital setting) can handle some of the basic maintenance tasks.

This could not only allow you some measure of immediate response should equipment suffer problems, reducing downtime, but it can also help to reduce the expense associated with ongoing equipment maintenance costs.

You just have to be aware of what you (or your support staff) are capable of handling so that you know when it's necessary to bring in professional help.

4. Hire Qualified Technicians

At the very least, you should have equipment serviced regularly by qualified technicians. Just like you hire an HVAC technician to inspect, clean, and maintain your home HVAC system annually and you take your car to the mechanic every three months or 3,000 miles, you should bring in the pros to maintain your medical equipment regularly.

You do have some options for service contracts, though. You could, for example, use OEM (original equipment manufacturer) technicians. These specialists are likely to be the costliest, but also the most qualified to maintain and repair equipment.

Another option is to hire third-party vendors. The upside here is that you'll likely pay a little less and technicians may be able to service a variety of equipment instead of just one brand or category. The potential drawback is that this vendor could be a Jack of all trades and a master of none, so to speak.

A final option is remote assistance. If you have an issue that requires immediate attention, customer support may be able to talk you through needed repairs or even access equipment that has internet connection capabilities in order to diagnose and fix problems.

5. Understand Warranties

Although proper usage, care, and maintenance should help to keep your equipment running properly for a good long while, problems can still occur. If this happens, you need to know how your warranty works so that you can receive compensation of some sort. Often, a failure to complete recommended maintenance could void your warranty.

General Principles of Care:

1. All objects to be disinfected or sterilized should first be thoroughly cleaned to remove all organic matter (blood and tissue) and other residue.

2. All items used in patient care shall be kept clean and in proper working condition.
3. All medical instruments and other items used for patient care must be cleaned and disinfected or sterilized before use on another patient.
4. Excessive moisture on electrical components can cause damage to certain instruments. Disinfectants and cleaners sprayed directly onto electrical devices may cause them to short circuit. This can also happen with excessively wet cleaning cloths.
5. Refer to the manufacturer's instructions for use of the appropriate type of disinfectant and the best method of cleaning and sterilizing each piece of medical instrumentation
6. Ethyl or isopropyl alcohol (70%-90%) is used for chemical disinfection.
7. For routine cleaning of most of the items, a ammonium germicidal detergent is available.
8. Cleaning and decontamination should begin as soon as possible after use as blood and body fluids can cause pitting of instruments and if left to dry can be difficult to remove.
9. When transport to the decontamination area is going to be delayed, soiled instruments should be moistened with a wet towel or enzymatic solution. 10. Decontamination and packaging should not be performed in patient care areas.
10. Perform periodic calibration of the equipment and the equipments that are faulty should be sent to the Bio-medical Engineering Department.
11. Clean properly
 - To prevent bacteria growth, the equipment should be inspected before each use and cleaned as often as necessary.
 - Use rubbing alcohol and a soft cloth or gauze pads to clean the equipment. If needed, dry the equipment with a clean paper towel and throw the towel away.
 - Also clean the equipment after each use to prevent buildup of debris.
12. Disinfect after each use
 - Besides keeping it clean, another important aspect of medical equipment maintenance is disinfecting after each use.
 - There are different disinfecting procedures depending on the equipment you are using. Check with your manufacturer or distributor for more information
13. Storage
 - Medical equipment should be stored in a clean and dry location so that it is not exposed to dust, direct sunlight, moisture, extreme temperatures or other potentially hazardous forces.

- Even when not in use medical equipment can deteriorate over time, make sure that you are regularly inspecting it. If you notice any damaged or defective parts, stop using the equipment and repair or replace the equipment
- Many types of medical equipment need to be serviced by qualified technicians at regular intervals. Make sure you are aware of this schedule

12 2.5 Learning Activities

2.5.1 Practical activity

Care and maintenance of a refrigerator.

You are provided with a refrigerator that has been in operation for one and half years without maintenance. Plan and execute its maintenance following the required principles.

2.5.2 Self-Assessment Questions

1. Enumerate most common examples of medical laboratory equipment used in the medical field.
2. Explain uses of medical imaging machines
3. List general principles of care for medical machines and equipment

2.5.3 Answers to Self-Assessment questions

1. Enumerate most common examples of medical laboratory equipment used in the medical field.

- Blood gas analyzers
- Chemistry analyzers
- Blood collection supplies
- Electrolyte analyzers
- Differential counters
- Drug testing analyzers
- Coagulation analyzers
- Hematology analyzers
- Urinalysis analyzers
- Microbiological systems

2. Explain uses of medical imaging machines

Medical imaging machines – Medical imaging is a type of technology that is used to create visual representation of the human body's interior. The visual image produced by the equipment is used for clinical analysis and medical intervention. There are many medical

imaging equipment, such as radiography (X-ray machine), computed tomography (CT scan), magnetic resonance imaging (MRI scan), ultrasound, and echocardiography, to name a few.

3. List general principles of care for medical machines and equipment

1. All objects to be disinfected or sterilized should first be thoroughly cleaned to remove all organic matter (blood and tissue) and other residue.
2. All items used in patient care shall be kept clean and in proper working condition.
3. All medical instruments and other items used for patient care must be cleaned and disinfected or sterilized before use on another patient.
4. Excessive moisture on electrical components can cause damage to certain instruments. Disinfectants and cleaners sprayed directly onto electrical devices may cause them to short circuit. This can also happen with excessively wet cleaning cloths.
5. Refer to the manufacturer's instructions for use of the appropriate type of disinfectant and the best method of cleaning and sterilizing each piece of medical instrumentation
6. Ethyl or isopropyl alcohol (70%-90%) is used for chemical disinfection.
7. For routine cleaning of most of the items, a ammonium germicidal detergent is available.
8. Cleaning and decontamination should begin as soon as possible after use as blood and body fluids can cause pitting of instruments and if left to dry can be difficult to remove.
9. When transport to the decontamination area is going to be delayed, soiled instruments should be moistened with a wet towel or enzymatic solution. 10. Decontamination and packaging should not be performed in patient care areas.
10. Perform periodic calibration of the equipment and the equipments that are faulty should be sent to the Bio-medical Engineering Department.
11. Clean properly
 - To prevent bacteria growth, the equipment should be inspected before each use and cleaned as often as necessary.
 - Use rubbing alcohol and a soft cloth or gauze pads to clean the equipment. If needed, dry the equipment with a clean paper towel and throw the towel away.
 - Also clean the equipment after each use to prevent buildup of debris.
12. Disinfect after each use

- Besides keeping it clean, another important aspect of medical equipment maintenance is disinfecting after each use.
- There are different disinfecting procedures depending on the equipment you are using. Check with your manufacturer or distributor for more information

13. Storage

- Medical equipment should be stored in a clean and dry location so that it is not exposed to dust, direct sunlight, moisture, extreme temperatures or other potentially hazardous forces.
- Even when not in use medical equipment can deteriorate over time, make sure that you are regularly inspecting it. If you notice any damaged or defective parts, stop using the equipment and repair or replace the equipment
- Many types of medical equipment need to be serviced by qualified technicians at regular intervals. Make sure you are aware of this schedule

13 2.6 Tools, Equipment, Supplies and Materials

The following resources are provided:

Tools

- PPEs
 - a. Safety boots
 - b. Goggles
 - c. Dust coats
 - d. First aid kit
 - e. Overalls
- Health facility
- Manufacturers' machine operating manuals for all laboratory equipment.
- Supplier's equipment warranty cards.
- Date of equipment purchase
- Running tap water
- Calibration reports
- Equipment maintenance schedules
- Laboratory practice manuals

- Stationery
- Cupboard drawers
- Internet

14 2.7 References

Kanyangarara, M., Munos, M. K., & Walker, N. (2017). Quality of antenatal care service provision in health facilities across sub-Saharan Africa: Evidence from nationally representative health facility assessments. *Journal of global health*, 7(2).

Ziraba, A. K., Mills, S., Madise, N., Saliku, T., & Fotso, J. C. (2009). The state of emergency obstetric care services in Nairobi informal settlements and environs: results from a maternity health facility survey. *BMC health services research*, 9(1), 1-8.